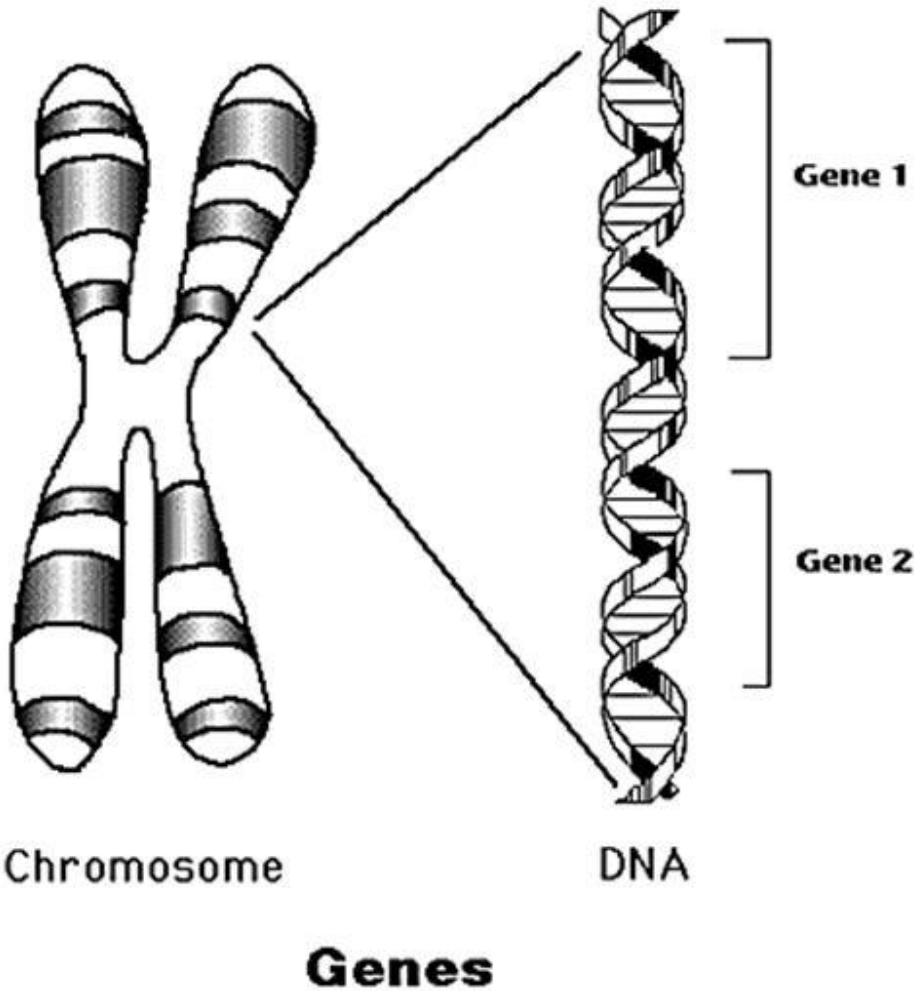
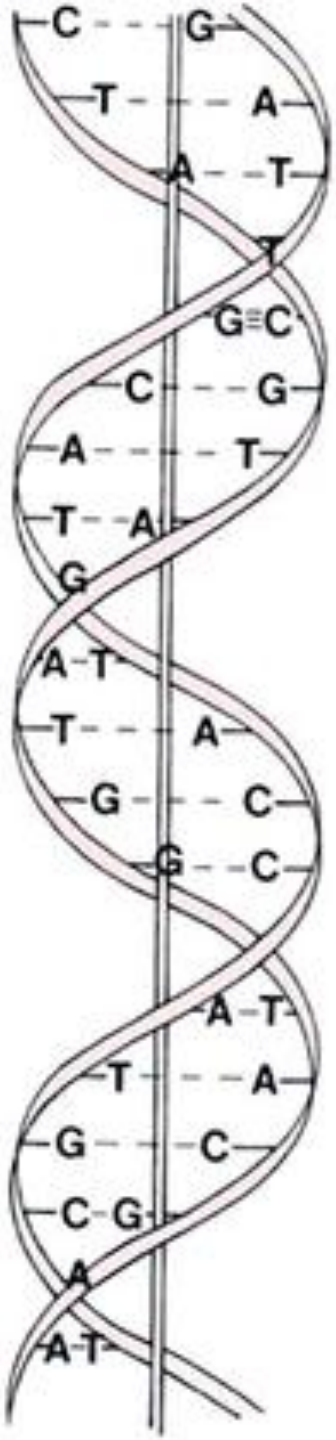
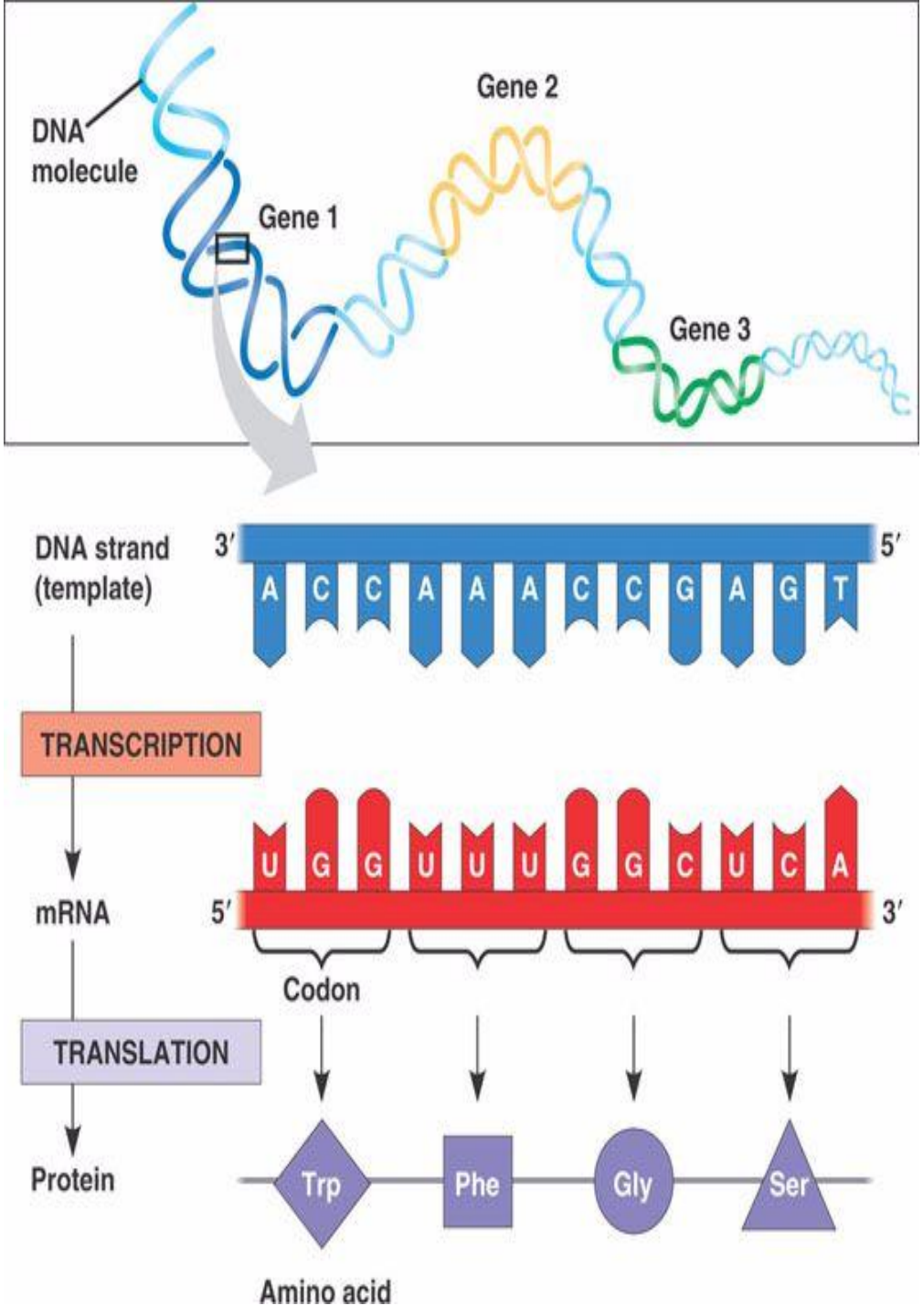
Types of mutations



Chromosomal mutation: affecting whole or a part of a chromosome

Gene mutation: changes to the bases in the DNA of one gene


C G
T A
A T
T
G C
C G
A T
T A
G
A T
T A
G C
G C
A T
T A
G C
C G
A
A T


DNA molecule
Gene 1
Gene 2
Gene 3
DNA strand (template)
3′
5′
A C C A A A C C G A G T
TRANSCRIPTION
mRNA
5′
3′
U G G U U U G G C U C A
Codon
TRANSLATION
Protein
Trp
Phe
Gly
Ser
Amino acid

A single chromosome contains thousands of genes, each encoding a protein.

TABLE 24–2 DNA, Gene, and Chromosome Content in Some Genomes

	Total DNA (bp)	Number of chromosomes*	Approximate number of genes
Escherichia coli K12 (bacterium)	4,639,675	1	4,435
Saccharomyces cerevisiae (yeast)	12,080,000	16†	5,860
Caenorhabditis elegans (nematode)	90,269,800	12‡	23,000
Arabidopsis thaliana (plant)	119,186,200	10	33,000
Drosophila melanogaster (fruit fly)	120,367,260	18	20,000
Oryza sativa (rice)	480,000,000	24	57,000
Mus musculus (mouse)	2,634,266,500	40	27,000
Homo sapiens (human)	3,070,128,600	46	29,000

GENETIC CODE TABLE:
Most genetic code tables designate the codons for amino acids as mRNA sequences

First letter	Second letter: U	Second letter: C	Second letter: A	Second letter: G	Third letter
U	UUU, UUC: Phe UUA, UUG: Leu	UCU, UCC, UCA, UCG: Ser	UAU, UAC: Tyr UAA: STOP UAG: STOP	UGU, UGC: Cys UGA: STOP UGG: Trp	U C A G
C	CUU, CUC, CUA, CUG: Leu	CCU, CCC, CCA, CCG: Pro	CAU, CAC: His CAA, CAG: Gln	CGU, CGC, CGA, CGG: Arg	U C A G
A	AUU, AUC, AUA: Ile AUG: Met	ACU, ACC, ACA, ACG: Thr	AAU, AAC: Asn AAA, AAG: Lys	AGU, AGC: Ser AGA, AGG: Arg	U C A G
G	GUU, GUC, GUA, GUG: Val	GCU, GCC, GCA, GCG: Ala	GAU, GAC: Asp GAA, GAG: Glu	GGT, GGC, GGA, GGG: Gly	U C A G

Key:

Ala = Alanine (**A**)
Arg = Arginine (**R**)
Asn = Asparagine (**N**)
Asp = Aspartate (**D**)
Cys = Cysteine (**C**)
Gln = Glutamine (**Q**)
Glu = Glutamate (**E**)
Gly = Glycine (**G**)
His = Histidine (**H**)
Ile = Isoleucine (**I**)
Leu = Leucine (**L**)
Lys = Lysine (**K**)
Met = Methionine (**M**)
Phe = Phenylalanine (**F**)
Pro = Proline (**P**)
Ser = Serine (**S**)
Thr = Threonine (**T**)
Trp = Tryptophan (**W**)
Tyr = Tyrosine (**Y**)
Val = Valine (**V**)

Type of code	Number of permutations
Singlet	$4^1 = 4$
Doublet	$4^2 = 16$
Triplet	$4^3 = 64$
Quadruplet	$4^4 = 256$
Pentuplet	$4^5 = 1024$

Only the triplet code really looks feasible

THE GENETIC CODE

- Each codon consists of three bases (triplet). There are 64 codons. They are all written in the 5' to 3' direction.
- 61 codons code for amino acids. The other three (UAA, UGA, UAG) are stop codons (or nonsense codons) that terminate translation.
- There is one start codon (initiation codon), AUG, coding for methionine. Protein synthesis begins with methionine (Met) in eukaryotes, and formylmethionine (fmet) in prokaryotes.
- The code is unambiguous. Each codon specifies no more than one amino acid.
- The code is degenerate. More than one codon can specify a single amino acid.
- All amino acids, except Met and tryptophan (Trp), have more than one codon.
- For those amino acids having more than one codon, the first two bases in the codon are usually the same. The base in the third position often varies.
- The code is almost universal (the same in all organisms). Some minor exceptions to this occur in mitochondria and some organisms.
- The code is commaless (contiguous). There are no spacers or "commas" between codons on an mRNA.

Gene Mutations

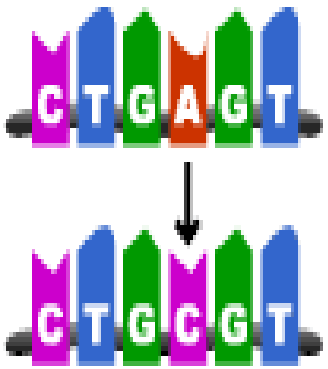
- ***Small scale:*** one gene is affected

- Any change to the DNA sequence of a gene:

  Nucleotides/Bases may be **added**, **missing**, or **changed**

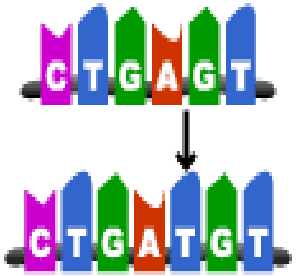
Gene Mutations: 2 Types

Point Mutation

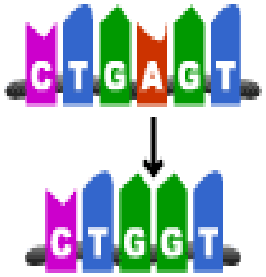


Point mutation

Frameshift Mutation



Insertion



Deletion

Point Mutation

- One base (A, T, C, or G) is **substituted for another**

Base pair substitutions- two types

a) Transitions

- Convert a purine-pyrimidine to the other purine-pyrimidine.
- 4 types of transitions; A $\leftrightarrow$ G and T $\leftrightarrow$ C

b) Transversions

- Convert a purine-pyrimidine to a pyrimidine-purine.
- 8 types of transversions; A $\leftrightarrow$ T, G $\leftrightarrow$ C, A $\leftrightarrow$ C, and G $\leftrightarrow$ T

- Case: ***Sickle-cell anemia***

- 3 Possible Consequences:
  - **nonsense mutations:** code for a stop, which can translate the protein
  - **missense mutations:** code for a different amino acid
  - **silent mutations:** code for the same amino acid

Frameshift Mutation

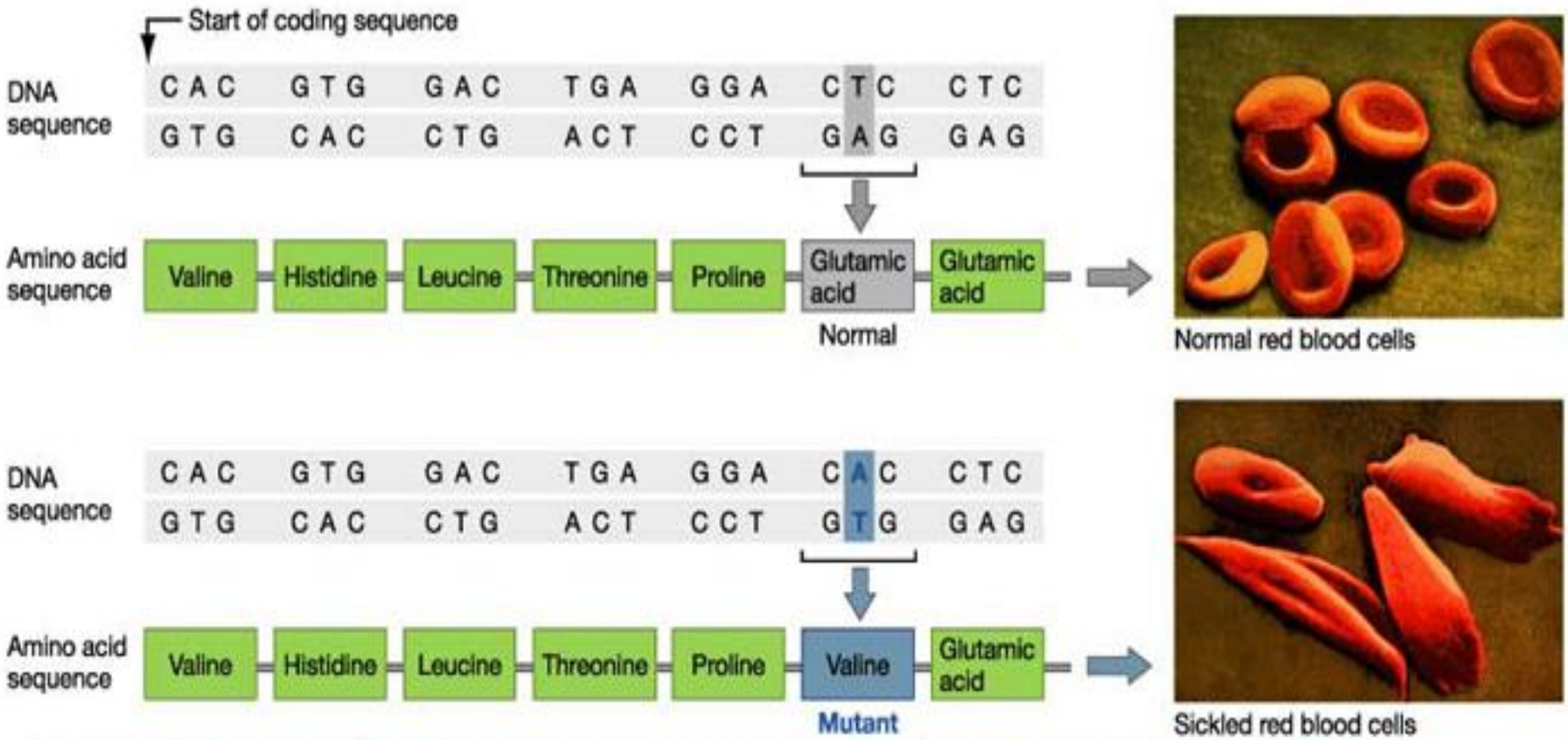
- One or more bases (A, T, C, or G) are added or deleted
- **Case: *Cystic Fibrosis***
- Caused by: Base pair deletions and insertions
  - **Insertion**: adding a base
  - **Deletion**: removing a base

Gene Mutations

- **KEY IDEA:** A mutated ***gene*** will make a mutated ***protein***

- **Mutant proteins are trouble!**
  - They do not ***go*** where they are supposed to go
  - They do not ***do*** what they are supposed to d

sickle-cell anemia



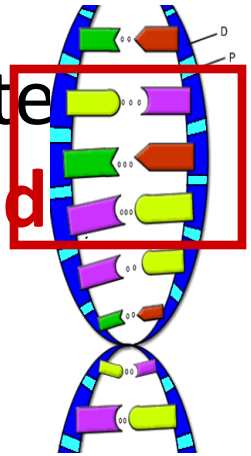
The change in amino acid sequence causes hemoglobin molecules to crystallize when oxygen levels in the blood are low. As a result, red blood cells sickle and get stuck in small blood vessels.

Example: Cystic Fibrosis (CF)

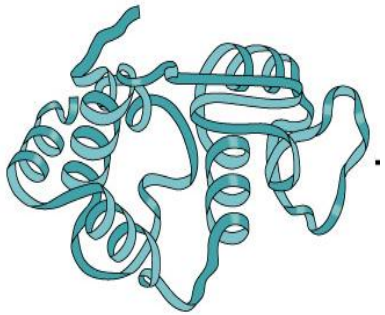
- ***What is it?***
  - **Autosomal, recessive disorder**
- ***Who gets it?***
  - **Anyone; most common in Caucasians**
- ***Prevalence***
  - **1 in ~3,000 Americans has CF**
  - **1 in 23 white Americans carries the mutant allele!**
- ***Symptoms***
  - **Thick mucus in the lungs and digestive track**
  - **Constant lung infections and impaired digestion**

Cause of Cystic Fibrosis (CF)

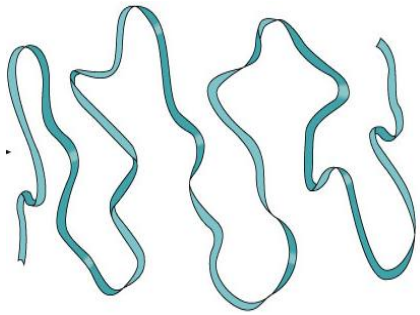
- The "CFTR" gene is mutate
  - 3 base pairs are **deleted**



- Mutant protein is missing an amino acid and cannot fold correctly



vs



KEY POINT

Mutation of a gene = Mutant protein

Dysfunctional proteins cause the symptoms of the disorder

Process of Mutations: Mutagenesis

a) Spontaneous: mainly during DNA Replication

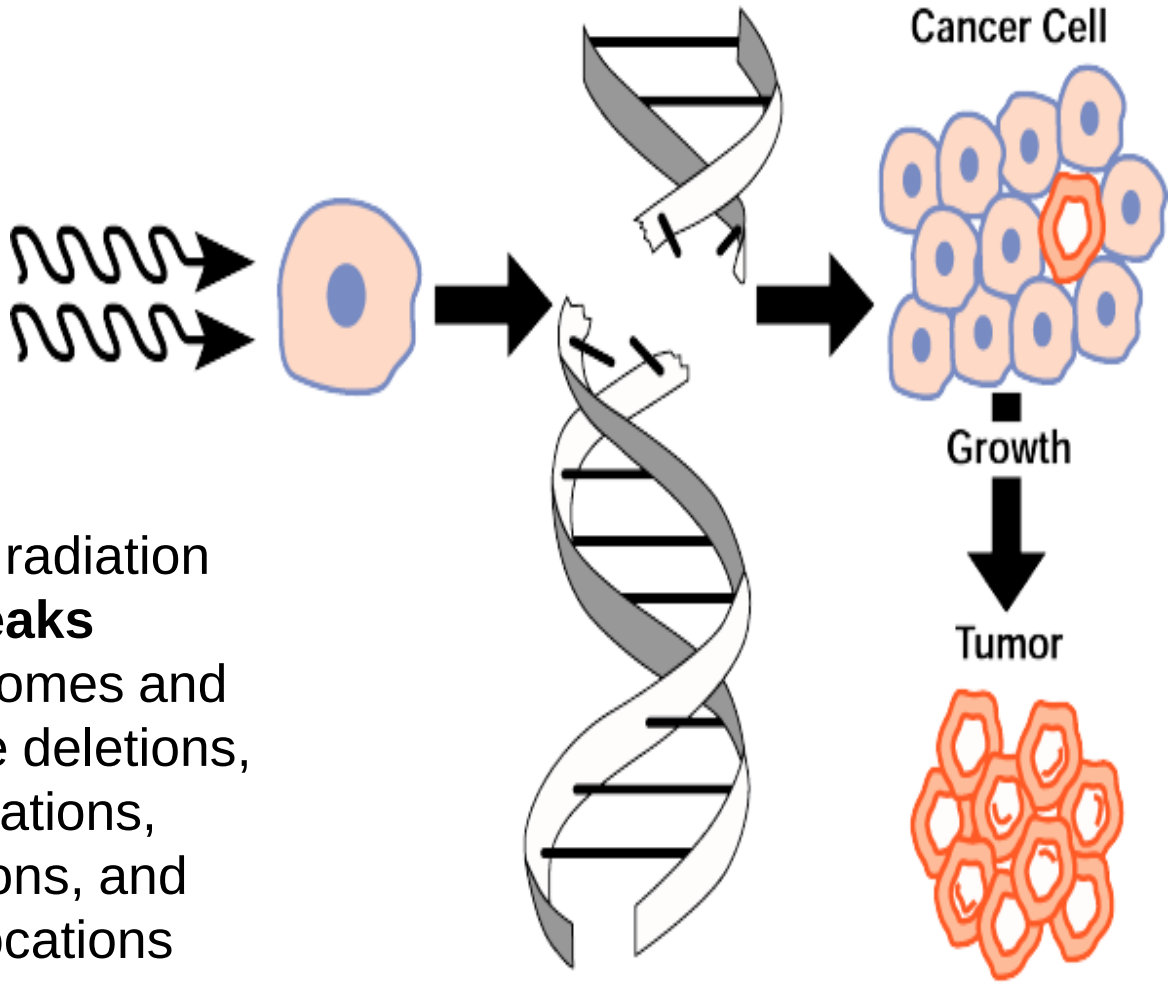
- **1 in 10^9 (a billion) replicated base pairs or 1 in 10^6 (a million) replicated genes. Mistakes occur during DNA Replication just before cell division. This is natural error rate of DNA polymerase.**
- **Many spontaneous errors are repaired.**

b) Induced mutations: Caused by MUTAGENS

- **Mutagens increase mistakes 10^{-5} (100 thousand) or 10^{-3} (a thousand) per replicated gene**

- **Physical mutagen: Ex: radiation from UV rays, X-rays or extreme heat**
- **Chemical mutagen: molecules that misplace base pairs or disrupt the helical shape of DNA.**

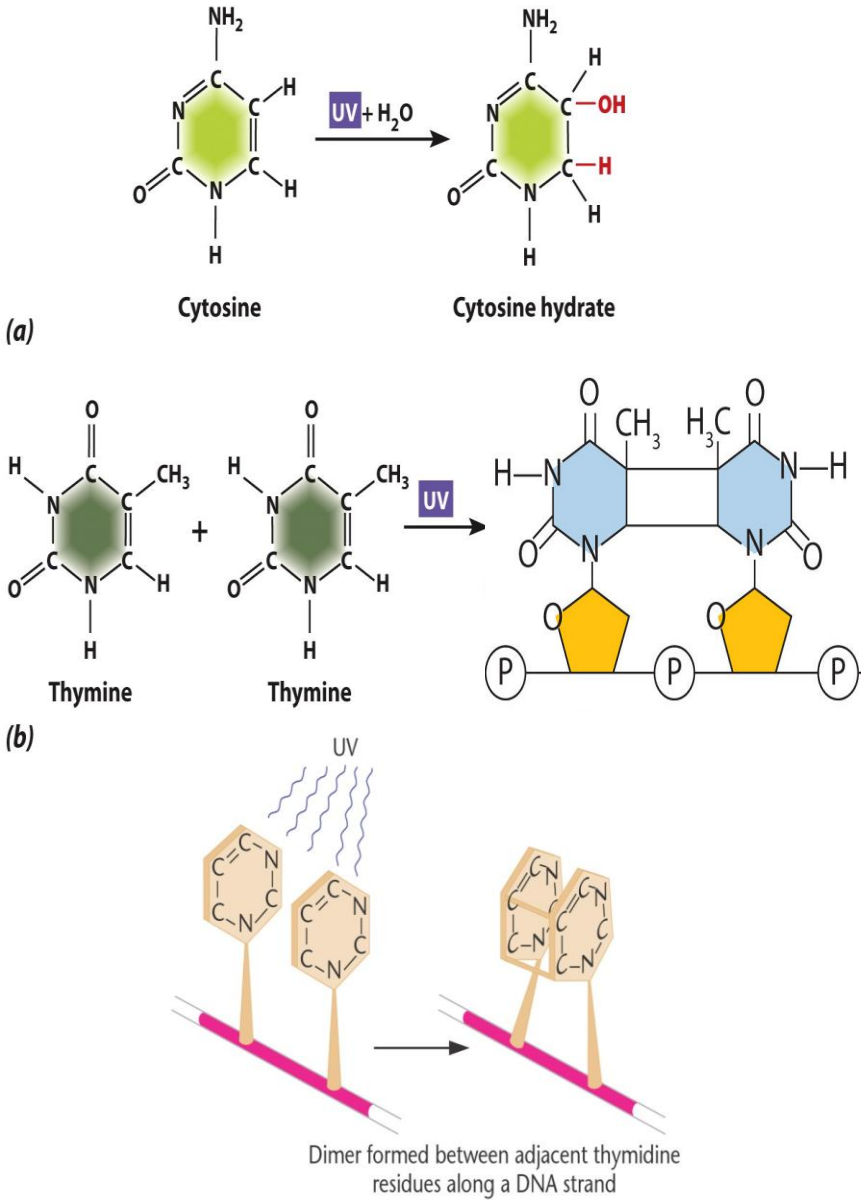
Physical Mutagen: X-rays and Gamma Rays Cause Breaks in DNA



Ionizing radiation **breaks** chromosomes and can cause deletions, duplications, inversions, and translocations

Figure 1. Development of cancer from mutation produced by ionizing radiation.

Physical Mutagen: Mutagenesis by Ultraviolet Irradiation



- Hydrolysis of cytosine to a hydrate may cause **mis-pairing** during replication
- Cross-linking of adjacent thymine forms **thymidine dimers**, which block DNA replication and activate DNA repair mechanisms.

Chemical Mutagens

Alkylating agents

$Cl—CH_2—CH_2—S—CH_2—CH_2—Cl$

Di-(2-chloroethyl) sulfide

(Mustard gas)

$CH_3—CH_2—O—SO_2—CH_3$

Ethyl methane sulfonate

(EMS)

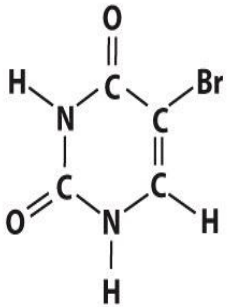
$CH_3—CH_2—O—SO_2—CH_2—CH_3$

Ethyl ethane sulfonate

(EES)

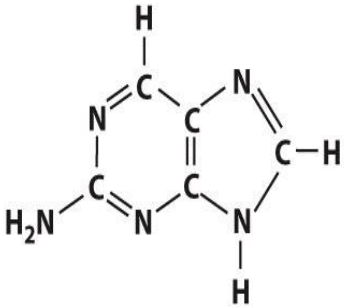
(a)

Base analogs



5-Bromouracil

(5-BU)

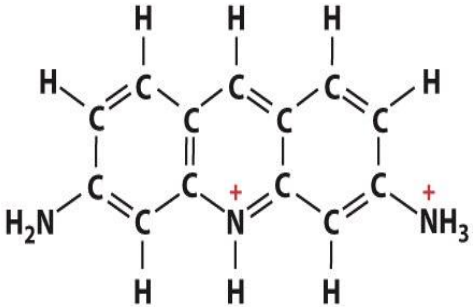


2-Aminopurine

(2-AP)

(b)

Acridines



2,8-Diamino acridine

(Proflavin)

(c)

Deaminating agent

HNO_2

Nitrous acid

(d)

Hydroxylating agent

NH_2OH

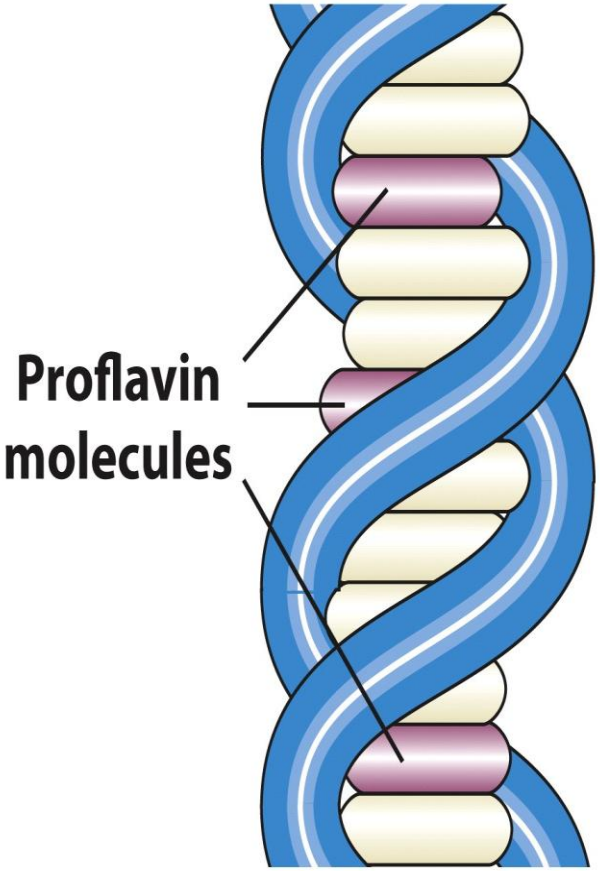
Hydroxylamine

(e)

Intercalation of an Acridine Dye Causes Frameshift Mutations

--(+) charges molecules

--Incorporated into DNA

--DNA is more rigid

--Change conformation (non-bending)